INCH-POUND

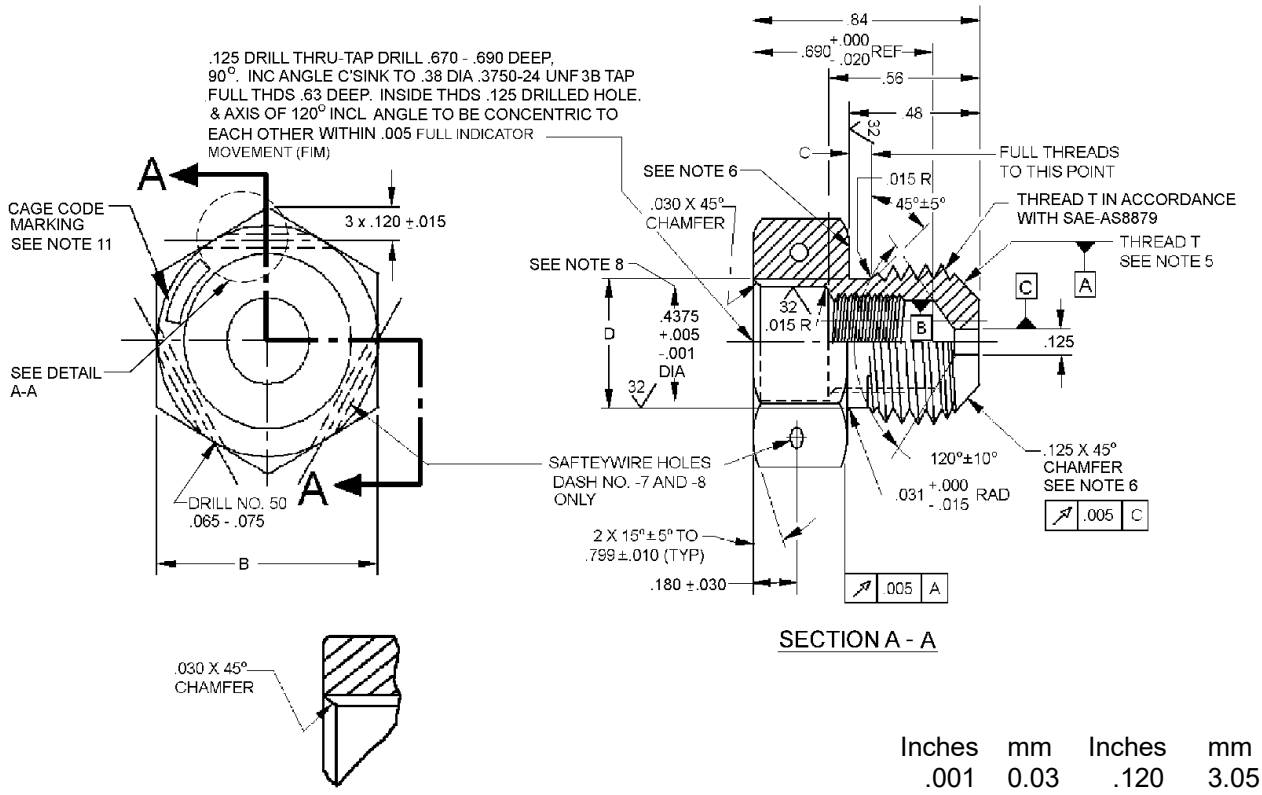
MS27612M
w/AMENDMENT 1
10 January 2020
SUPERSEDING
MS27612M
1 August 2013

DETAIL SPECIFICATION SHEET

ADAPTER, HYDRAULIC BLEEDER VALVE, AIRCRAFT WHEEL BRAKE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



Inches	mm	Inches	mm
.001	0.03	.120	3.05
.005	0.13	.125	3.18
.006	0.15	.180	4.57
.010	0.25	.4375	11.113
.015	0.38	.48	12.2
.020	0.51	.56	14.2
.030	0.76	.63	16.0
.031	0.79	.690	17.5
.065	1.65	.799	20.29
.075	1.91	.84	21.3

FIGURE 1. Adapter dimensions and configuration.

AMSC N/A

FSC 4820

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Part or Identifying Number (PIN) (see note 9)	T Thread Size (see note 7)	B +.003 -.004 (+0.08) (-0.10) inch (mm)	C inch (mm)	D Dia. +.002 -.003 (+0.05) (-.0.08) inch (mm) (see note 5)
MS27612-5	.5625-18UNJF-3A	.8130 (20.650)	.083 +.015 -.000 (2.11 +0.38 -0.00)	.481 (12.22)
MS27612-7				
MS27612-6	.7500-16UNJF-3A	1.000 (25.40)	.100 +.009 -.006 (2.54 +0.23 -0.15)	.660 (16.76)
MS27612-8				

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Dimensioning and tolerancing in accordance with ASME Y14.5. Unless otherwise specified, tolerances for decimals ± .005 inch (0.13 mm), angles ±.5°.
4. Surface texture: Symbols in accordance with ASME Y14.36, requirements in accordance with ASME B46.1. Unless otherwise specified, surfaces to be 150 μin R_a, except as noted.
5. "D" diameter and outside thread "T" shall be concentric within .005 inch (0.13 mm) FIM.
6. This surface to be a smooth conical surface free from burrs, tool marks and visible flat spots. Annular tool marks will be allowed to 100 μin R_a maximum.
7. Male threads in accordance with SAE-AS8879.
8. .4375 inch (11.113 mm) DIA and inside thread shall be concentric within .002 inch (0.05 mm) FIM.
9. MS27612-7 and -8 adapters with safety wire holes are preferred for new design.
10. Remove sharp edges, burr and slivers.
11. Marking of part shall be any permanent marking method that will not damage the part. The minimum character height shall be 1/16 inch (.0625 inch) (1.57 mm).

FIGURE 1. Adapter dimensions and configuration - Continued.

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REQUIREMENTS

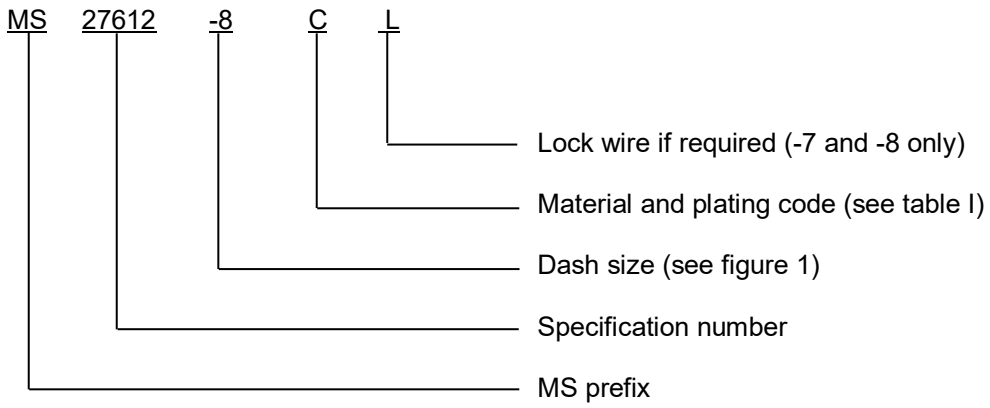
Material and plating requirements and codes, see table I.

TABLE I. Materials and plating's.

Material and plating code	Material/plating
C	Carbon steel bar in accordance with ASTM A108 (1117 or 12L14) or ASTM A576, (grade 1117 UNS G11170). Cadmium plating in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick.
CZ	Carbon steel bar in accordance with ASTM A108 or ASTM A576, (grade 1117 UNS G11170 or 12L14). Zinc plating in accordance with ASTM B633, type VI, FeZn 25.
F	Alloy steel bar in accordance with or SAE-AMS6370 (type 4130, UNS G41300) or SAE-AMS-S-6758 (SAE 4130), SAE-AMS6349 (SAE 4140, UNS G41400) or SAE-AMS6382 (SAE 4140, UNS G41400). Cadmium plating in accordance with SAE-AMS-QQ-P-416, type II, class 3, 200µ inches to 300µ inches (5.08 µm to 7.62 µm) thick.
FZ	Alloy steel bar in accordance with or SAE-AMS6370 (type 4130, UNS G41300) or SAE-AMS-S-6758 (SAE 4130), SAE-AMS6349 (SAE 4140, UNS G41400) or SAE-AMS6382 (SAE 4140, UNS G41400). Zinc plating in accordance with ASTM B633, type VI, FeZn 25.
S	Corrosion resistant steel in accordance with SAE-AMS5659, SAE-AMS5862, ASTM A564/A564M (UNS S15500 or type XM-12) or SAE-AMS5665 (UNS N06600). Passivate in accordance with SAE-AMS2700, type 6 or 7.

Platings. All platings shall be capable of meeting the 96 hour salt spray test in accordance with ASTM B117 Fluid passages, other openings, and internal threads shall not be subject to the plating thickness requirement and may have bare areas provided they are protected with a light film of oil.

PIN example:



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Guidance on use of alternative parts with less hazardous or non-hazardous materials. This specification provides an alternate material, corrosion resistant steel, and zinc finish via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

Marking: Part shall be permanently marked with the MS PIN, and include the manufacturers CAGE.

Table II provides a detailed cross-reference of superseded MS27612 PINs, new MS27612 PINs, and preferred MS27612 PINs.

TABLE II. Cross-reference data.

Superseded MS PIN	New MS PIN	Preferred MS PIN
MS27612-5	MS27612-5C	MS27612-7CL
MS27612-6	MS27612-6C	MS27612-8CL
MS27612-7	MS27612-7CL	MS27612-7CL
MS27612-8	MS27612-8CL	MS27612-8CL

Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Referenced documents shall be of the issue in effect on date of invitations for bid.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. This document references the following:

ASME B46.1	SAE-AMS-QQ-P-416
ASME Y14.5	SAE-AMS2700
ASME Y14.36	SAE-AMS5659
ASTM A108	SAE-AMS5665
ASTM A564/A564M	SAE-AMS5862
ASTM A576	SAE-AMS6349
ASTM B117	SAE-AMS6370
ASTM B633	SAE-AMS6382
SAE-AMS-S-6758	SAE-AS8879

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 70
DLA - CC

Preparing activity:
DLA-CC

(Project 4820-2020-001)

Review activities:
Army - AT
Navy - MC, SA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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